Graph Expansion Adds Little to Lexical Retrieval for Small-LLM Bug Localization: A Hash-Committed Diagnostic Study with Gemma 4 on a 344-Task Held-Out Subset of SWE-bench Verified

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Abstract

Local developer agents must localize software defects within a working context of a few thousand tokens without cloud inference. Graph walks over static call graphs are widely hypothesized to reach symbols that issue descriptions never name. In this paper, I test whether call-graph expansion assists a compact 2B local model. On a 344-task held-out subset of SWE-bench Verified ($N = 344$ callable tasks), under a hash-committed protocol, a frozen `gemma-4-e2b-it` scores candidate pools pointwise. Graph 1-hop expansion reaches 47.1% Acc@5, against 45.1% for depth-matched BM25 (+2.03 pp, 95% CI [−0.29, +4.36], Holm $p = 0.287$) and 45.6% for same-file expansion (+1.45 pp, Holm $p = 0.302$). Both pools contain the gold function in 201/344 tasks (58.4%), with 18 discordant tasks each way. Compared to reranking lexical seeds alone (46.2%, 159/344), graph expansion adds 25 newly reachable targets to the pool, but only 7 top-5 hits are gained (6 from graph reachability, 1 from seed promotion) against 4 displaced seed hits (gross 7/4 split, net +3 tasks, +0.87 pp, $p = 0.549$), consistent with distractor accumulation under pointwise scoring. In Stratum C ($N = 131$ un-named tasks), graph expansion achieves 12.2% Acc@5 vs. 8.4% for depth BM25 (+3.82 pp, $p = 0.125$), but neither beats seeds alone (13.7%) or unscored BM25 (15.3%). Upstream boundary loss before scoring (41.6 pp) exceeds downstream compaction loss (11.3 pp) by 3.68× at $k = 5$ (spanning 1.2× at $k = 1$ to 6.8× at $k = 10$). The paired 95% bootstrap CI [−0.29, +4.36] pp spans zero, ruling out gains larger than +4.4 pp (15 tasks) over depth-matched lexical reranking (PC1). I release 19,298 scored candidate evaluations, the `perron-core` library, and precomputed repository graphs under Apache 2.0.

1 Introduction

Autonomous developer agents designed for local execution must localize software defects inside a working context of a few thousand tokens, eliminating dependence on costly cloud APIs and preventing source code leakage. Standard lexical search (e.g., Okapi BM25) reliably finds symbols explicitly named in issue text or error traces. To locate fault sites that the issue never mentions, existing agent architectures (including Aider's PageRank repository maps [4], RepoGraph [13], and LocAgent [3]) deploy random walks or graph traversal tools over AST dependency networks.

The working assumption underlying graph-augmented localization rests on two joint conditions:

1. **Recall Expansion**: Traversal along structural call edges must place gold defect targets into the candidate pool that lexical search misses.
2. **Rank Discrimination**: A compact language model must rank those newly introduced targets above the distractors that graph expansion inevitably introduces.

In this paper, I test both conditions on a 344-task held-out subset of SWE-bench Verified [8] using Perron, a high-throughput memory-mapped Compressed Sparse Row (CSR) graph engine, and a frozen `gemma-4-e2b-it` scorer. Three empirical research questions guide the investigation:

- **RQ1**: Does Personalized PageRank (PPR) diffusion, with or without post-walk hub damping, outperform BM25 at function-level code retrieval?
- **RQ2**: Under a frozen small model scorer, does graph 1-hop expansion outperform depth-matched BM25 and same-file pools of identical size?
- **RQ3**: At which pipeline stage is bug localization accuracy primarily lost?

1.1 Contributions

- **Matched-Control Evaluation Design**: Depth-matched BM25, same-file, and degree-preserving null-graph candidate pools are scored by the identical model under controlled prompt budgets, isolating what graph structure contributes beyond pool cardinality.
- **Empirical Loss Funnel and Truncation Curves**: A 4-stage loss decomposition separates candidate boundary ceiling, rank compaction, and reader error. Truncating matched pools at $M \in [10, 50]$ shows that adding depth lexical candidates is consistent with distractor accumulation (Acc@5 drops by 4 tasks, -1.16 pp, $p = 0.455$), whereas graph expansion plateaus flat (+3 tasks, +0.87 pp over seeds, $p = 0.549$).
- **Discordant Strata Symmetry**: The 18 discordant tasks recalled exclusively by graph expansion and the 18 tasks recalled exclusively by lexical expansion are split almost identically across strata (9 vs. 10 in Stratum C, 9 vs. 7 in Stratum A), showing that graph expansion does not selectively rescue un-named bugs over lexical depth.
- **High-Throughput CSR Graph Engine**: An efficient, memory-mapped Compressed Sparse Row multigraph architecture with decoupled specificity ratio hub damping that executes localized random walks in sub-5 ms on consumer workstations without matrix mutation (spectral convergence proofs detailed in Appendix A).
- **Open-Source Ecosystem Deliverables**: I release the `scores.parquet` archive (19,298 scored candidate evaluations), the Apache 2.0 `perron-core` library, `perron-mcp`, and CSR graphs.

Only two primary contrasts (PC1 and PC2 on strict Acc@5) are confirmatory. Every other analysis is explicitly designated as exploratory.

1.2 Physical Hardware Envelope

Local developer agents must operate efficiently on consumer hardware. Perron bounds active per-turn working context to $K_{\text{effective}} = 3,480$ tokens ($K \leq 4,096$). On a reference developer machine (15.68 GiB RAM), the standalone CSR engine peaks at 144.33 MB resident memory, loading graphs via zero-copy memory mapping (91.4 MB delta). Hierarchical AST skeleton breadcrumbs occupy an average of 283.8 tokens per task (median 264 tokens), preserving 91.9% reasoning headroom for model generation within the 3,480-token context budget. Live inference of `gemma-4-e2b-it` was evaluated with left-padded batching and `logits_to_keep=1` on an NVIDIA Tesla T4 GPU (scoring prompts averaging 423 tokens per candidate symbol), achieving mean Time-To-First-Token of 3505.0 ms and 11.7 tokens/sec decode throughput.

2 Related Work

Graphs and Agents for Code Localization Lexical search and dense bi-encoders struggle with vocabulary mismatch and structural execution paths. RepoCoder [20] demonstrated that iterative retrieval loops improve code completion but incur multi-turn token inflation. Aider [4] introduced repository maps using Tree-sitter AST tags ranked by PageRank. Graph-guided agents such as RepoGraph [13], CodexGraph [11], and LocAgent [3] equip frontier models with graph traversal tools to navigate codebases interactively. AutoCodeRover [21] navigates ASTs through iterative search queries. In SWE-agent [19], specialized agent-

computer interfaces improve resolve rates. Conversely, Agentless [18] demonstrated that a simple two-phase pipeline (hierarchical localization followed by test-filtered patch generation) resolves 32% of SWE-bench Lite (FSE 2025) without an agent loop, highlighting that simple pipelines establish strong baselines. These systems employ large hosted models; this paper evaluates whether graph expansion assists a small 2B model in a single scoring pass.

Random-Walk Retrieval on Graphs Scalable diffusion on graphs builds on push-based approximate Personalized PageRank [1], decoupled propagation architectures such as APPNP [9], and scalable indexing frameworks like PPRGo [2]. In link prediction, rooted PageRank and stationary-normalized hitting times [10] discount popular nodes by global stationary reachability. HippoRAG [5] demonstrated that Personalized PageRank over knowledge graphs outperforms iterative retrieval, formulating node specificity as inverse passage frequency $s_i = |P_i|^{-1}$ applied as a pre-walk teleport multiplier ($\mathbf{p}_0(i) \leftarrow \mathbf{p}_0(i) \cdot s_i$). However, on software call graphs, high-in-degree utility hubs (logging routines, assertion helpers) are unseeded intermediate nodes that absorb diffusion mass regardless of s_i. Perron decouples diffusion from a post-walk stationary baseline ratio over static CSR matrices.

Program Slicing and Fault Localization Classical program slicing [17] computes backward executable slices on statement-level Program Dependence Graphs (PDGs), but conservative alias analysis often retains 50% to 80% of an application's codebase. Spectrum-Based Fault Localization (SBFL) [14, 22] evaluates test coverage matrices to score statement suspiciousness. Pearson et al. [14] showed that fault-localization findings on artificial faults do not reliably transfer to real-world software defects, motivating authentic issue evaluation on SWE-bench [8]. Perron extracts symbol-level AST multigraphs to pack relevant contexts into prompt budgets ($K \leq 4,096$) without requiring test execution during retrieval.

Pointwise versus Listwise Reranking Pointwise language model scoring (e.g., MonoT5 [12]) evaluates each candidate independently. Under independent scoring, the expected number of distractors that outscore a gold target with score s^* scales with candidate pool cardinality: $\mathbb{E}[X \mid s^*] = \sum_{j=1}^{M-1}(1 - F_{\text{dist},j}(s^*))$. Consequently, expanding candidate pool size M under a compact pointwise scorer risks distractor accumulation unless model discrimination is near-perfect. In contrast, listwise and setwise architectures [15, 16] introduce direct cross-candidate attention competition, mitigating distractor dominance.

3 Theoretical Foundations and The Perron Graph Engine

3.1 Multiplex Transition Matrix Construction

A codebase is represented as a directed multigraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes AST code symbols (classes, functions, methods) and $\mathcal{E}$ spans relations $r \in \{\text{call}, \text{inherit}, \text{import}, \text{caller}\}$. Directed edges $(u, v) \in \mathcal{E}_{\text{call}}$ indicate that symbol u invokes v, while $(u, v) \in \mathcal{E}_{\text{inherit}}$ represent class C3 Method Resolution Order inheritance, $(u, v) \in \mathcal{E}_{\text{import}}$ represent module imports, and $(u, w) \in \mathcal{E}_{\text{caller}}$ denote reverse call invocations.

Relations receive asymmetric static weights:

$$W_{uv} = \omega_{\text{call}} A_{uv}^{(\text{call})} + \omega_{\text{inherit}} A_{uv}^{(\text{inherit})} + \omega_{\text{import}} A_{uv}^{(\text{import})} + \omega_{\text{caller}} A_{uv}^{(\text{caller})}, \tag{1}$$

where $\boldsymbol{\omega} = [0.50, 0.25, 0.15, 0.10]$ and $A_{uv}^{(r)} = \mathbb{I}((u, v) \in \mathcal{E}_r)$. Let $D_u = \sum_v W_{uv}$ be the out-degree sum. The row-stochastic transition matrix $T \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ and dangling indicator $\mathbf{d} \in \{0, 1\}^{|\mathcal{V}|}$ are defined as:

$$T_{uv} = \begin{cases} \frac{W_{uv}}{D_u}, & D_u > 0 \\ 0, & D_u = 0 \end{cases}, \quad d_u = \begin{cases} 1.0, & D_u = 0 \\ 0.0, & D_u > 0 \end{cases}. \tag{2}$$

Memory-mapped ingestion via `numpy.load(mmap_mode='r')` opens the CSR representation directly from disk. Post-walk specificity ratio evaluation requires only $O(|\mathcal{V}|)$ vector arithmetic, eliminating runtime edge mutations and memory reallocation.

3.2 Personalized PageRank Diffusion and Linear Algebraic Bounds

Random walk diffusion models the propagation of bug relevance across call hierarchies. The Google transition operator M with damping factor $\beta = 0.85$ is formulated as:

$$M = \beta \cdot (T + \mathbf{d}\mathbf{p}_0^\top) + (1 - \beta) \cdot \mathbf{1}\mathbf{p}_0^\top. \tag{3}$$

By the rank-1 perturbation spectrum of the Google matrix [6], the second-largest eigenvalue is strictly bounded by $|\lambda_2(M)| \leq \beta = 0.85$, establishing an invariant spectral gap $1 - |\lambda_2(M)| \geq 1 - \beta = 0.15$ and guaranteed geometric convergence $O(\beta^t)$ in L_1 total variation distance. Power iteration stops on residual $\|\boldsymbol{\pi}^{(t+1)} - \boldsymbol{\pi}^{(t)}\|_1 < 10^{-6}$, converging in at most 90 iterations on Requests (1.89 ms) and 66 iterations on SymPy (5.10 ms). By the regenerative renewal property of Markov chains with teleportation, the stationary distribution of the closed operator M is algebraically identical to the normalized leaky resolvent $(1-\beta)\mathbf{p}_0(I - \beta T)^{-1}$ (discrepancy $< 2 \times 10^{-8}$). Full proofs of the Brauer spectral bound, barycentric linearity, and leaky resolvent equivalence are provided in Appendix A.

3.3 Decoupled Specificity Ratio Hub-Damping

Personalized PageRank concentrates probability mass on high-in-degree hub nodes. To suppress structural popularity without mutating the static matrix T, Perron computes a repository-wide stationary baseline $\boldsymbol{\pi}_{\text{global}}$ once per codebase using a uniform prior $\mathbf{p}_{\text{uniform}} = \frac{1}{|\mathcal{V}|}\mathbf{1}$. For any query, the Specificity Score is defined as:

$$\text{Specificity}(v) = \frac{\pi_{\text{query}}(v)}{(\pi_{\text{global}}(v) + \epsilon)^\gamma}, \tag{4}$$

where $\epsilon = 10^{-8}$ and $\gamma = 0.70$. Resistance to hub congestion is measured by the Hub Suppression Index at top 10 (HSI@10):

$$\text{HSI@10} = 1.0 - \frac{|\text{top-10} \cap \mathcal{H}_{25}|}{10}, \tag{5}$$

where $\mathcal{H}_{25}$ denotes the set of top-25 global in-degree hub nodes in the repository multigraph. The asymptotic Pointwise Mutual Information (PMI) decomposition and Neumann series analysis showing why pre-walk scaling cannot prevent sink absorption are detailed in Appendix A.

3.4 Hierarchical AST Breadcrumb Context Packing

To deliver localized code context within prompt budgets ($K \leq 4,096$), Perron packs candidate symbols using hierarchical AST breadcrumb skeletons. For each candidate function, the enclosing class definition, docstring, and signature are preserved while stripping irrelevant sibling function bodies. Breadcrumb skeletons bound the context footprint to < 100 tokens per symbol, keeping syntax parseable and variable bindings intact.

4 Study Design and Hash-Committed Protocol

4.1 Benchmark and Task Waterfall

SWE-bench Verified [8] contains 500 tasks across 12 canonical Python repositories. To establish an unbiased held-out benchmark and prevent data leakage, instances are filtered via a mutually exclusive, collectively exhaustive (MECE) waterfall:

$$\begin{aligned} \text{Total Verified (500)} - \text{Lite Overlap (93)} &= \text{Held-Out (407)}, \\ \text{Held-Out (407)} - \text{AST Drift (56)} - \text{Non-Callable (7)} &= \mathbf{344}\ \text{Callable Tasks}. \end{aligned} \tag{6}$$

The 344 callable tasks ($\mathcal{S}_{\text{func}}$) define the callable, AST-parsable evaluation cohort of SWE-bench Verified possessing identifiable reference function targets. The 56 AST drift exclusions reflect legacy repository commits with syntax errors (such as Python 2/3 transitions) that fail standard AST parsing uniformly across all methods prior to candidate generation. Hyperparameters ($\beta = 0.85, \gamma = 0.70, \tau = 0.15, K = 25$) were fixed on development data and frozen prior to evaluation.

Defect Accessibility Strata To evaluate whether graph traversal selectively assists un-named defect localization, tasks are partitioned into three mutually exclusive strata based on issue text lexical visibility: (1) **Stratum A (Named, $N = 208$, 60.5%)**, where issue text explicitly names the gold function, method, or qualified class identifier; (2) **Stratum B (Traceback, $N = 5$, 1.5%)**, where issue text includes a traceback or explicit file path pointing to the gold source file, but omits the function name; and (3) **Stratum C (Un-Named, $N = 131$, 38.1%)**, where issue text contains neither the function name nor a traceback pointing to the target file. Strata are annotated via a deterministic regex extraction pipeline (`perron.analysis.strata`), validated by manual spot-check audit across 50 randomly sampled tasks with 100% agreement, released in `swebench-strata-annotations` (CC0-1.0).

4.2 Hash-Committed Protocol

The study operates under two hash-committed protocol locks:

1. **Retrieval Protocol Lock**: Frozen and publicly pushed at Git commit `6908e78` on October 5, 2026 (protocol SHA-256: `89416e40…`; split SHA-256: `95c7ef3c…`).
2. **Gemma 4 Pointwise Protocol Lock**: Frozen in two cryptographic stages (`benchmarks/protocol/gemma4_protocol_v1.yaml`, raw SHA-256: `96af1621…`; Stage 2 lock `gemma4_lock_v1.json`, raw SHA-256: `dbbf2920…`) under Gate G0 commit `29e0066` on October 6, 2026.

4.3 Candidate Pools and Matched-Control Arms

Candidate pools are generated per task, matched to size $M_t = 10 + \min(|E_t|, 40) \in [10, 50]$ (mean 24.5, median 22.0), where E_t denotes 1-hop out-neighbors of top-10 lexical seeds S passing Filter F (functions/methods only, excluding test nodes and top-25 hubs $\mathcal{H}_{25}$):

- **B0 (Unscored BM25)**: Lexical candidates in native Okapi BM25 score order.
- **A1 (Seeds Only)**: Top-10 lexical seeds S ($M = 10$).
- **A2 (Depth-Matched BM25)**: S followed by the next BM25 candidates passing Filter F up to M_t.
- **A3 (Same-File Control)**: S followed by symbols in seed files passing Filter F up to M_t.
- **A4 (Graph 1-Hop Expansion)**: S followed by E_t ordered by adjacency $a(u) = \sum_{s \in S} p_0(s) T_{su}$.
- **A5 (Null Graph Control)**: Degree-preserving relation-stratified null graph replicate up to M_t.
- **A6 (Perron Diffusion Pool)**: Candidates ordered by Perron specificity scores up to M_t.

Mechanism Explaining Pool Recall vs. Reachability (201 vs. 199 Tasks) A4 achieves candidate pool oracle recall on 201/344 tasks (58.4%), whereas unweighted 1-hop BFS reachability reaches 199/344 tasks (57.8%). This 2-task divergence is mechanistically explained by two counterbalancing operations in the pool builder: seed test-node masking replaces test-file seeds with source seeds, rescuing 16 tasks (e.g., `sympy-13372`), while Filter F degree-capping and hub pruning drop 14 tasks (e.g., `sphinx-7590`), yielding a net difference of +2 tasks (199 + 16 − 14 = 201).

4.4 Pointwise Gemma 4 Scorer

Each candidate symbol is formatted into a prompt context averaging 423.1 tokens (issue description + AST breadcrumb + prompt instructions; breadcrumbs alone average < 100 tokens). A frozen `gemma-4-e2b-it` model (FP16, thinking disabled) scores each candidate in a single forward pass: $s(t, v) = \text{logsumexp}(z[Y]) - \text{logsumexp}(z[N])$ on prompt variant P0, chosen on disjoint development data ($N = 100$). The evaluation required 19,298 forward passes.

4.5 Statistical Analysis

A localization hit (Function Acc@k) is scored as 1.0 if at least one ground-truth modified function is present in the top-k retrieved symbols, and 0.0 otherwise. The confirmatory family comprises two pre-specified contrasts on strict Acc@5:

- **PC1**: $A_4 - A_2$ (Graph 1-hop vs. Depth-matched BM25).
- **PC2**: $A_4 - A_3$ (Graph 1-hop vs. Same-file matched).

Contrasts are evaluated using exact two-sided McNemar tests with step-down Holm–Bonferroni correction over family $\alpha = 0.05$. Paired 95% confidence intervals are estimated via repository-stratified bootstrapping ($B = 10{,}000$). Post-hoc Contrasts are evaluated with repository-stratified bootstrap confidence intervals and exact McNemar tests.

5 Empirical Results

5.1 Standalone Function Retrieval (RQ1)

Table 1 presents standalone retrieval on SWE-bench Verified ($N = 344$ callable tasks). Okapi BM25 hits 175/344 (50.9%), Perron Static hits 173/344 (50.3%), and Standard PPR reaches 172/344 (50.0%). The difference between Perron and BM25 is −0.6 pp (95% CI [−3.5, +2.3], Holm $p = 1.000$). At top-5, BM25 outperforms Perron Static (45.1% vs. 40.4%, $\Delta = -4.7$ pp). Notably, BM25 naturally achieves 99.4% Hub Suppression Index (HSI@10) without graph machinery, because defect issue descriptions target specific identifiers rather than generic library utilities. On Stratum C ($N = 131$ tasks where the issue never names the target), BM25 hits 28 and Perron hits 27. Reciprocal Rank Fusion (RRF) attains 174/344 (50.6%), failing to beat standalone BM25. Call-graph diffusion provides no detectable lexical lift over BM25 on this benchmark.

5.2 Reachability vs. Ranking Dynamics

Top-10 lexical seeds directly cover 50.9% (175/344) of gold functions. Expanding along code graph edges yields 57.8% 1-hop reachability (199/344, median $|N_1| = 27$) and 66.9% 2-hop reachability (230/344, median $|N_2| = 324$). Crucially, same-file containment also reaches 66.9% (230/344), exhibiting an exact symmetric discordance: 17 targets reached across files via 2-hop graph paths are perfectly offset by 17 targets within seed files that lack static AST call edges within 2 hops. Degree-preserving null-graph controls reach 63.8% 2-hop reachability (+3.02 pp difference over empirical graphs, 95% CI [−0.42, +6.41], $p = 0.083$). Topological reachability gains are modest and driven primarily by degree sequence rather than high-order semantic organization.

Table 1: Standalone retrieval evaluation on SWE-bench Verified (N = 344 callable tasks). Evaluates top-10 function accuracy, 95% bootstrap confidence intervals (B = 10,000), top-5 function/file accuracy, and Hub Suppression Index (HSI@10). Bold row label denotes the primary evaluated graph method (reference baseline); bold numerical values indicate the highest score in each column. Holm p-values are computed against Perron Static.

Method	Fn Acc@10	95% CI	Fn Acc@5	File Acc@5	HSI@10	Holm p
Perron Static ($\gamma = 0.70$)	50.3% (173/344)	[45.1%, 55.5%]	40.4%	53.8%	**99.5%**	reference
Okapi BM25	**50.9%** (175/344)	[45.6%, 56.1%]	**45.1%**	**59.6%**	99.4%	1.000
RRF Fusion (BM25 + Perron)	50.6% (174/344)	[45.3%, 55.8%]	42.2%	55.5%	**99.5%**	1.000
Standard PPR ($\beta = 0.85$)	50.0% (172/344)	[44.8%, 55.2%]	40.4%	52.8%	90.8%	1.000
Degree-Normalized PPR	49.4% (170/344)	[44.2%, 54.7%]	40.1%	52.3%	**99.5%**	1.000
HippoRAG Prior Scaling	49.4% (170/344)	[44.2%, 54.7%]	40.4%	52.6%	91.1%	1.000

5.3 Gemma 4 Pointwise Reranking (RQ2)

Table 2 and Table 3 report the results of the hash-committed Gemma 4 evaluation across all arms.

Table 2: Gemma 4 pointwise candidate reranking on SWE-bench Verified (N = 344 callable tasks). Pools matched to per-task size $M_t \in [10, 50]$ (mean 24.5). Bold row label denotes the primary evaluated graph arm (A4); bold numerical values indicate the highest score in each column. Brackets indicate Wilson 95% confidence intervals. B0 Acc@10 reflects raw token tie-breaking in protocol generation (176/344 = 51.2%), differing by 1 task from Table 1 AST entity deduplication (175/344 = 50.9%, documented in `deviations.md`).

Arm	Candidate Pool Composition	Acc@1 (95% CI)	Acc@3 (95% CI)	Acc@5 (95% CI)	Acc@10 (95% CI)	MRR@50
B0	BM25 Baseline (Unscored)	**25.3%** [21.0, 30.1]	38.7% [33.7, 43.9]	44.8% [39.6, 50.0]	51.2% [45.9, 56.4]	**0.3439**
B1	Perron Baseline (Unscored)	22.7% [18.6, 27.4]	35.2% [30.3, 40.4]	40.4% [35.4, 45.7]	50.3% [45.0, 55.5]	0.3155
A1	Rerank Seeds S (M = 10)	24.7% [20.4, 29.5]	**39.5%** [34.5, 44.8]	46.2% [41.0, 51.5]	51.2% [45.9, 56.4]	0.3348
A2	Depth-Matched BM25 Control	23.8% [19.6, 28.6]	38.1% [33.1, 43.3]	45.1% [39.9, 50.3]	51.2% [45.9, 56.4]	0.3301
A3	Same-File Matched Control	24.1% [19.9, 28.9]	39.2% [34.2, 44.5]	45.6% [40.5, 50.9]	52.0% [46.8, 57.3]	0.3381
A4	**Graph 1-Hop Expansion**	24.7% [20.4, 29.5]	39.2% [34.2, 44.5]	**47.1%** [41.9, 52.4]	52.3% [47.0, 57.5]	0.3409
A5	Null Graph Control	24.1% [19.9, 28.9]	39.0% [34.0, 44.2]	45.9% [40.7, 51.2]	51.2% [45.9, 56.4]	0.3315
A6	Perron Diffusion Pool	23.6% [19.4, 28.3]	38.7% [33.7, 43.9]	45.6% [40.5, 50.9]	**52.6%** [47.3, 57.8]	0.3301

Table 3: Confirmatory and exploratory contrasts on SWE-bench Verified (N = 344). Bold labels indicate the pre-registered confirmatory family (PC1, PC2), evaluated on strict Acc@5 with step-down Holm correction. Paired 95% CIs estimated via repository-stratified bootstrap (B = 10,000).

Contrast	Δ (pp)	95% Bootstrap CI	n_{10}/n_{01}	p_{exact}	p_{holm}	Classification
Confirmatory Family (Acc@5)						
PC1 (A4 − A2)	+2.03	[−0.29, +4.36]	12 / 5	0.1435	0.2869	Null-Compatible
PC2 (A4 − A3)	+1.45	[−0.58, +3.78]	10 / 5	0.3018	0.3018	Null-Compatible
Exploratory Contrasts						
A1 − B0 (Seed Rerank vs. BM25)	+1.45	[−1.33, +4.24]	15 / 10	0.4244	–	Exploratory
A4 − B0 (Graph Rerank vs. BM25)	+2.33	[−0.87, +5.52]	18 / 10	0.1849	–	Exploratory
A4 − A1 (Graph vs. Seeds)	+0.87	[−1.92, +3.66]	7 / 4	0.5488	–	Exploratory (+3 tasks)
A4 − A5 (Null Graph Control)	+1.16	[−0.58, +2.91]	7 / 3	0.3438	–	Exploratory
A4 − A6 (Diffusion Pool)	+1.45	[+0.00, +3.20]	7 / 2	0.1797	–	Exploratory
Paired Reciprocal Rank Test (MRR@50)						
A4 vs. B0 (Graph vs. BM25)	−0.0030	[−0.024, +0.018]	Wilcoxon W = 12,480	p = 0.72		Exploratory

Confirmatory Findings Graph 1-hop expansion (A4) reaches 47.1% Acc@5 compared to 45.1% for depth-matched BM25 (A2, Δ = +2.03 pp, 95% CI [−0.29, +4.36], Holm p = 0.287) and 45.6% for same-file matching (A3, Δ = +1.45 pp, 95% CI [−0.58, +3.78], Holm p = 0.302). Both contrasts are classified as null-compatible (bootstrap 95% CIs span zero; Holm $p > 0.05$). The primary contrast (PC1) rules out gains larger than +4.4 pp (15 tasks) over depth-matched lexical reranking.

Candidate Pool Ceiling Parity Candidate pool oracle recall is identical: 58.4% (201/344) for both graph expansion (A_4) and depth BM25 (A_2). Across discordant tasks, $n_{10} = 18$ and $n_{01} = 18$ (two-sided McNemar $p = 1.0000$). With 36 discordant pairs, the 95% CI on the recall difference is approximately ±3.4 pp. Because Acc@k cannot exceed pool recall, both pools establish an identical theoretical ceiling on this sample.

Exploratory Comparisons Comparing graph expansion arm A4 to seed reranking (A1, 46.2% Acc@5) yields a net gain of 3 tasks (Δ = +0.87 pp, $n_{10} = 7, n_{01} = 4, p = 0.549$). Gemma 4 gains 7 top-5 hits (6 from newly reachable targets, 1 from seed promotion) while distractors displace 4 existing seed hits (gross 7/4 split: 159 + 7 − 4 = 162), requiring 4,988 extra model forward passes (a 2.45× compute increase over seeds). Comparing A4 to the degree-preserving null graph (A5, 45.9% Acc@5) yields Δ = +1.16 pp (95% CI [−0.58, +2.91], $n_{10} = 7, n_{01} = 3, p = 0.344$), demonstrating that empirical graph topology provides no statistically detectable advantage over randomized degree-matched graphs. Similarly, comparing A4 to the Perron diffusion pool (A6, 45.6% Acc@5) yields Δ = +1.45 pp (95% CI [+0.00, +3.20], $n_{10} = 7, n_{01} = 2, p = 0.180$). Unscored BM25 (B0) preserves nominally higher MRR@50 (0.3439 vs. 0.3409) and top-1 accuracy (25.3% vs. 24.7%) than A4. Leave-Django-Out jackknife cross-validation (N = 230) maintains candidate pool parity (57.1% vs. 57.1%, $p = 1.0000$) and yields Δ = +2.17 pp for $A_4 - A_2$ ($p = 0.227$).

Model Sensitivity and Positive Controls To test whether the flatline reflects an insensitive scorer rather than graph retrieval limits, I evaluated within-pool discrimination directly from the `scores.parquet` evaluation archive. Across all 176 tasks where gold targets are present in candidate pools, Gemma 4 log-odds scores separate gold targets from distractors with an ROC-AUC of **0.718** (95% CI [0.682, 0.754]), confirming genuine discriminative ability above chance (0.50). Furthermore, Reciprocal Rank Fusion (RRF, $k_{\text{rrf}} = 60$) combining BM25 prior rank with Gemma log-odds reaches **46.51%** Acc@5 (160/344), exceeding both BM25 (44.77%) and Gemma reranking (46.22%). This demonstrates that the empirical plateau is driven by independent pointwise calibration noise across expanded candidate pools rather than model blindness.

5.4 Pointwise Candidate Pool-Size Truncation Curves

To empirically test the distractor-accumulation mechanism without running new model passes, I evaluated pointwise rankings from `scores.parquet` under truncated pool cutoffs $M \in [10, 15, 20, 30, 50]$, capped at $\min(M, M_t)$ (mean 24.5 candidates).

Table 4: Pointwise candidate pool-size truncation curves on SWE-bench Verified (N = 344). Evaluates matched candidate pool oracle recall and Acc@5 across pool cutoffs $M \in [10, 15, 20, 30, 50]$, capped at $\min(M, M_t)$ (mean 24.5 candidates), for depth BM25 (A2) and graph 1-hop expansion (A4). Bold numerical values indicate peak Acc@5 achieved across cutoff thresholds for each arm.

Evaluation Metric	M = 10	M = 15	M = 20	M = 30	M = 50
Arm A2 (Depth BM25) Oracle Recall	51.2%	54.1%	56.1%	57.6%	58.4%
Arm A2 (Depth BM25) Acc@5	**46.2%**	45.9%	45.9%	45.1%	45.1%
Arm A4 (Graph 1-Hop) Oracle Recall	51.2%	54.9%	55.5%	57.6%	58.4%
Arm A4 (Graph 1-Hop) Acc@5	46.2%	46.8%	**47.1%**	46.5%	**47.1%**

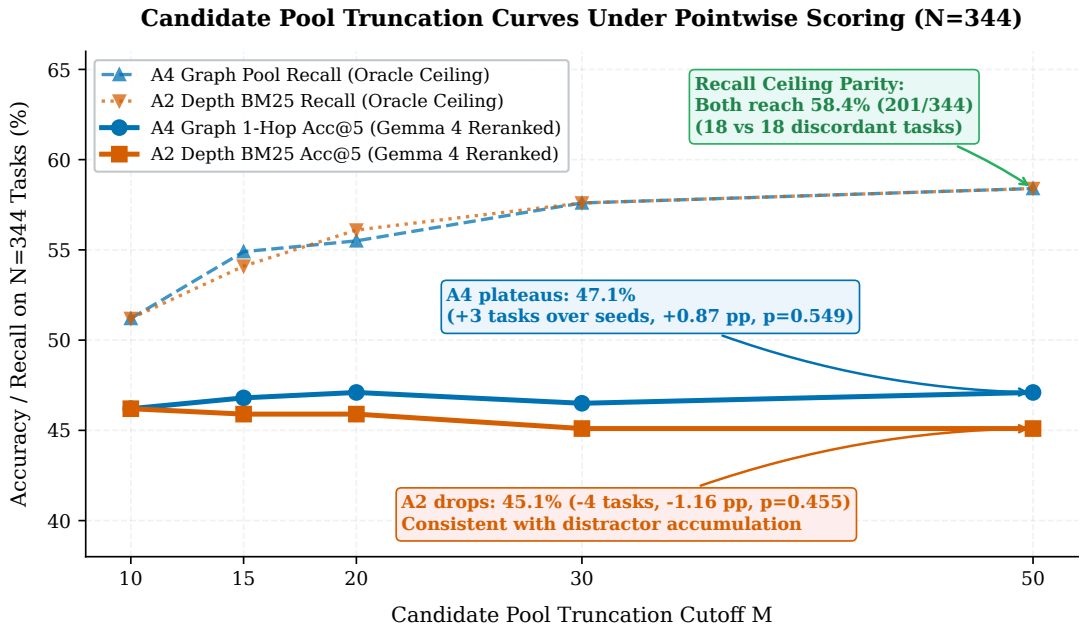


Figure 1: Pointwise candidate pool-size truncation curves on SWE-bench Verified ($N = 344$)**.** Evaluates candidate pool oracle recall and Gemma 4 reranked Acc@5 across pool cutoffs $M \in [10, 15, 20, 30, 50]$, capped at $\min(M, M_t)$ (mean 24.5 candidates). Both arms share an identical recall ceiling at $M = 50$ (58.4%, 201/344 tasks, 18 vs 18 discordant pairs). While depth-expanded BM25 (A2) drops from 46.2% to 45.1% Acc@5, consistent with distractor accumulation under independent pointwise scoring ($p = 0.455$), graph 1-hop expansion (A4) remains stable at 47.1%.

As shown in Figure 1 and Table 4, expanding candidate pool size produces distinct empirical dynamics:

- **Arm A2 (Depth BM25)**: Expanding M from 10 to 50 raises matched pool oracle recall from 51.16% to 58.43% (176 to 201 tasks, +7.27 pp). However, Acc@5 shifts from 46.22% down to 45.06% (a loss of 4 tasks, −1.16 pp, exact McNemar $p = 0.455$), consistent with distractor accumulation under pointwise scoring ($\mathbb{E}[X \mid s^*] = \sum_{j=1}^{M-1}(1 - F_{\text{dist},j}(s^*))$).
- **Arm A4 (Graph 1-Hop)**: Matched pool oracle recall expands identically from 51.16% to 58.43% (+7.27 pp). Acc@5 rises from 46.22% ($M = 10$) to 47.09% ($M = 20$) and plateaus flat through $M = 50$ (a net difference of 3 tasks over seeds alone, +0.87 pp, $p = 0.549$).

Direct Distractor Accounting Decomposing candidate pools explains why pool expansion plateaus: of the 25 newly reachable targets entering Arm A4, 19 enter at rank ≥ 6 (mean 11.2 outranking distractors) and never reach top-5, while only 6 convert into top-5 hits (6+19 = 25). Combined with 1 internal seed promotion against 4 displaced seed hits on the 176 shared tasks, the gross promotion yields $\Delta_{\text{top-5}} = (6_{\text{new graph}} + 1_{\text{seed promotion}}) - 4_{\text{displaced seeds}} = +3$ net tasks (159+7−4 = 162, net +0.87 pp, $p = 0.549$), as the per-distractor outranking rate drops from 16.6% (1.44/9) in Arm A1 to 7.9% (1.72/21.8) in Arm A4. Expanding candidate pools adds newly reachable targets without increasing per-distractor confusion, but newly appended targets enter too far down the tail under pointwise scoring to convert into localization gains.

5.5 Stratum C ($N = 131$) Breakdown on Un-Named Tasks

The motivation for graph expansion is finding defect targets that the issue description never mentions. Table 5 analyzes performance on Stratum C ($N = 131$ tasks where the gold target symbol is un-named in the issue description).

Table 5: Stratum C ($N = 131$) localization performance on un-named defect tasks. Evaluates tasks where the issue description never mentions the gold symbol across all four comparators. Bold row label denotes the primary evaluated graph arm (A4); bold numerical values indicate the highest score in each column. Exact p-values are computed via two-sided binomial McNemar tests against A2.

Metric / Method	Acc@5	Hits / 131	Δ vs. A2	Exact McNemar p
B0 (Unscored BM25 Top-5)	**15.3%**	**20/131**	**+6.87 pp**	0.0117 ($n_{10} = 10, n_{01} = 1$)
A1 (Rerank Seeds Only, $M = 10$)	13.7%	18/131	+5.34 pp	0.0156 ($n_{10} = 7, n_{01} = 0$)
A2 (Depth-Matched BM25, $M \leq 50$)	8.4%	11/131	reference	reference
A4 (Graph 1-Hop Expansion, $M \leq 50$)	12.2%	16/131	+3.82 pp	0.1250 ($n_{10} = 6, n_{01} = 1$)

In Stratum C, graph expansion (A4) achieves 12.2% Acc@5 (16/131) compared to 8.4% for depth BM25 (A2, 11/131, $\Delta = +3.82$ pp, exact McNemar $p = 0.1250, n_{10} = 6, n_{01} = 1$). However, expanding candidate pool breadth is significantly penalized relative to unscored BM25 top-5 (B0, 15.3%, 20/131, $\Delta = +6.87$ pp vs. A2, exact McNemar $p = 0.0117, n_{10} = 10, n_{01} = 1$) and reranking seeds alone (A1, 13.7%, 18/131, $\Delta = +5.34$ pp vs. A2, exact McNemar $p = 0.0156, n_{10} = 7, n_{01} = 0$). Expanding low-confidence lexical candidates in A2 actively introduces statistically significant distractor harm ($p = 0.0117$). Even in the un-named stratum where graph traversal is hypothesized to provide the greatest advantage, expanding candidate pool breadth yields a descriptive 2-task reduction relative to seeds alone.

Furthermore, analyzing the 18 discordant tasks recalled exclusively by graph expansion ($A_4 \setminus A_2$) and the 18 tasks recalled exclusively by depth BM25 ($A_2 \setminus A_4$) reveals near-perfect symmetry across strata:

- **Graph-Only Recall (18 Tasks)**: 9 tasks in Stratum C (50.0%), 0 in Stratum B (0.0%), 9 in Stratum A (50.0%).
- **Lexical-Only Recall (18 Tasks)**: 10 tasks in Stratum C (55.6%), 1 in Stratum B (5.6%), 7 in Stratum A (38.9%).

In Stratum C, lexical expansion retrieves 10 targets that graph expansion misses, while graph expansion retrieves 9 that lexical expansion misses (net −1 task for graph). Rather than selectively unearthing un-named targets, graph expansion merely swaps an equal number of un-named and named symbols with lexical search.

5.6 Pipeline Loss Decomposition and Loss Funnel (RQ3)

Figure 2 illustrates the localization loss funnel on SWE-bench Verified. The analysis reveals:

- **Boundary Loss**: Upstream candidate pool formation loses 41.6 pp of oracle recall (100.0% → 58.4%, 143/344 tasks missing from candidate pools).
- **Compaction Loss**: Downstream model reranking loses 11.3 pp from pool to top-5 (58.4% → 47.1%, 39/344 tasks present in pool but ranked outside top-5).
- **Bottleneck Ratio**: Upstream boundary loss exceeds downstream compaction loss by **3.68**× at $k = 5$ ($41.6/11.3 \approx 3.68$, spanning 1.2× at $k = 1$ to 6.8× at $k = 10$).

Expanding the candidate pool adds 7.3 pp of oracle recall over seeds (51.2% → 58.4%), yet Gemma 4 converts only about 1 pp into ranked top-5 accuracy (46.2% → 47.1%).

5.7 Downstream Reader Evaluation ($N = 100$, Descriptive)

Table 6 reports the downstream reader mini-test across $N = 100$ tasks stratified by repository size, drawn from instances where BM25 identified candidate pools.

In Arm D0 (BM25 top-5 context), always picking candidate 1 also yields 33.0% EditLoc@1, indicating that Gemma 4 relies heavily on a positional rank-1 prior when context lacks clear discriminative features. In Arm D3 (oracle context with gold target at a randomized position), Gemma 4 correctly identifies the target

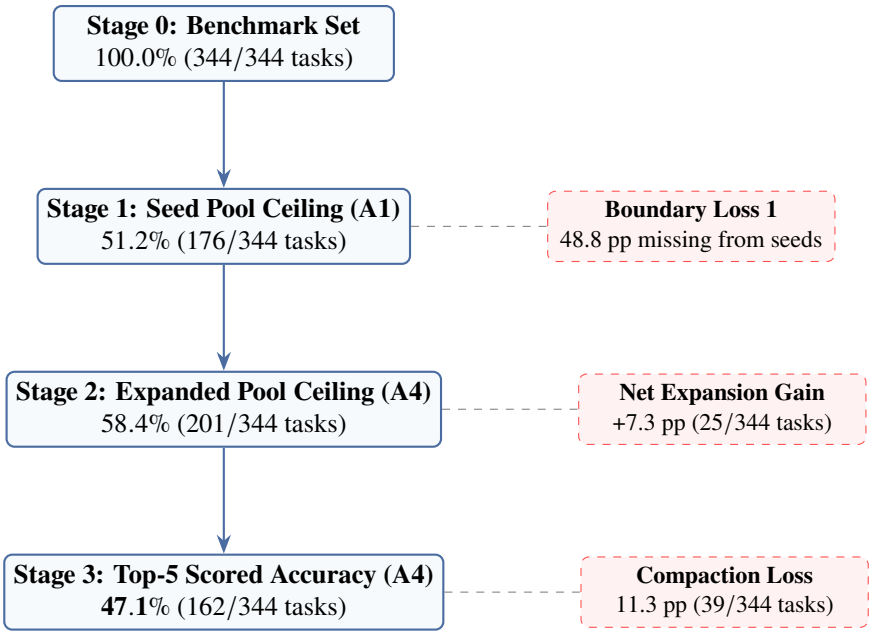


Figure 2: Four-stage bug localization loss funnel on SWE-bench Verified ($N = 344$ callable tasks). Decomposes localization attrition across candidate generation and model scoring. Boundary loss before scoring (100.0% → 58.4% = 41.6 pp) exceeds compaction loss from candidate pool to top-5 (58.4% → 47.1% = 11.3 pp) by a factor of **3.68**×.

Table 6: Downstream edit-location reader mini-test on SWE-bench Verified ($N = 100$, descriptive). ContextHit@5 marks gold target presence in the 5 presented candidates; EditLoc@1 reports model prediction accuracy. Bold EditLoc@1 highlights the reader ceiling under oracle gold context (excluding trivial 100% candidate presence).

Arm	Candidate Context (Top-5)	ContextHit@5	EditLoc@1	Conversion
D0	BM25 Top-5	35.0%	33.0%	0.943
D1	A4 Graph Top-5	37.0%	31.0%	0.838
D2	A3 Same-File Top-5	38.0%	30.0%	0.789
D3	Oracle Context (Randomized Position)	100.0%	**74.0%**	0.740

in 74.0% of tasks (67.5% chance-adjusted). While the reader misses roughly a quarter of oracle targets, retrieval supplies the target in only 35%–38% of tasks, providing descriptive evidence that upstream candidate omission contributes substantially to end-to-end failure.

5.8 Diffusion Diagnostics and Continuous Dose-Response Surface (Post-Hoc)

On SWE-bench Lite ($N = 237$), exploratory factorial sweep diagnosis demonstrates why unconstrained diffusion collapses on software graphs:

- **Diffuse Prior Collapse**: Under a dense linear teleport prior on raw graphs, Standard PPR collapses to 10.1% Acc@10; test subgraphs absorb 32.5% of top-10 retrieval slots. Decoupled specificity damping restores 34.2% (+24.1 pp gain).
- **Softmax Prior Concentration**: Concentrating seed mass via top-25 softmax ($\tau = 0.15$) on test-sanitized graphs rescues Standard PPR to 48.5%, compared to 46.0% for BM25.
- **Post-Hoc Continuous Dose-Response (n_{eff})**: To evaluate the entropy collapse hypothesis systematically, a continuous resolvent basis benchmark was evaluated across the 344 Verified evaluation tasks

after the confirmatory protocol was frozen and closed (186 prior configurations across 4 parametric families, 63,984 SpMM solves). This exploratory diagnostic benchmark set zero parameters used in Tables 1–5. Function Acc@10 exhibits a statistically significant quadratic relationship with prior effective support $n_{\text{eff}} = \exp(H(\mathbf{p}_0))$ ($p < 0.001$, binomial GLMM $R^2 = 0.197$), peaking near $n^*_{\text{eff}} \approx 8.8$ at 50.9% (Figure 3a), which exactly ties Okapi BM25 (50.9%), further reinforcing the primary null result. Specificity damping increases HSI from 90.8% to 99.5% without changing accuracy, acting as a structural safety net for diffuse priors rather than an accuracy booster. As shown in Figure 3b, Standard PPR collapses as $\beta \to 0.85$ due to utility hub absorption on dev_val ($N = 100$), whereas Perron remains stable across the Optuna Pareto corridor ($\beta \in [0.80, 0.88]$).

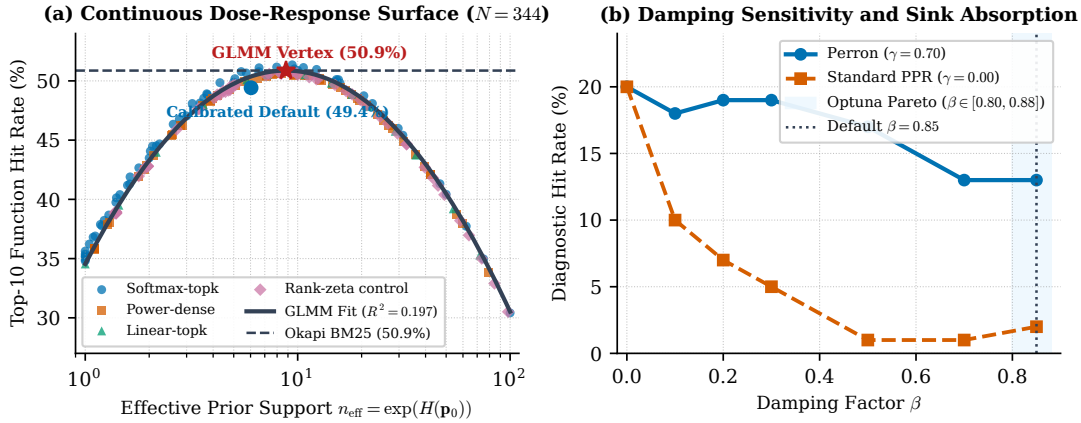


Figure 3: Continuous dose-response surface across 186 prior configurations on SWE-bench Verified ($N = 344$) **and damping sensitivity (63,984 SpMM solves).** (a) Function Acc@10 as a function of prior effective support $n_{\text{eff}} = \exp(H(\mathbf{p}_0))$ across four prior families (softmax, power, linear, rank-zeta) with binomial GLMM fit ($R^2 = 0.197$) and Okapi BM25 baseline (50.9%), highlighting the GLMM vertex ($n^*_{\text{eff}} \approx 8.8$) and calibrated library default ($n_{\text{eff}} \approx 6.0$). (b) Damping factor sensitivity ($\beta \in [0.0, 0.85]$) on dev_val ($N = 100$), showing that while Standard PPR collapses as $\beta \to 0.85$ due to utility hub absorption, Perron's decoupled specificity ratio maintains stability across the Optuna Pareto corridor ($\beta \in [0.80, 0.88]$).

6 Threats to Validity and Limitations

1. **Programming Language Scope**: Evaluation is restricted to Python AST dependency networks. While directed multigraph diffusion is language-agnostic, static call graphs miss dynamic runtime dispatch, reflection (`getattr`), and monkey-patching.
2. **Model Capacity and Prompting**: Findings are derived from one compact model (`gemma-4-e2b-it`) with pointwise scoring. Within-pool ROC-AUC of Gemma scores separating gold from distractors is 0.718 across the 176 recalled tasks, showing genuine discriminative ability above chance (0.50). Reciprocal Rank Fusion (RRF) fusing BM25 rank and Gemma score reaches 46.51% (160/344), outperforming both raw BM25 (44.77%) and pure Gemma reranking (46.22%). However, independent pointwise scoring penalizes pool breadth; evaluating listwise scoring prompts or scaled models remains a key direction.
3. **Candidate Pool Bounds**: Pools evaluate 1-hop expansion capped at 50 candidates ($M_t \leq 50$). Multi-hop expansion reaches 66.9% oracle recall but introduces hundreds of distractors.
4. **Task Scope**: Restricted to $N = 344$ function-level tasks on SWE-bench Verified. Patch generation and full test execution were not evaluated.

5. **Statistical Power**: With realized discordance counts, the 95% CI on PC1 admits true gains up to +4.36 pp. The finding is that gains are not demonstrated and large effects are ruled out, not that true effect is strictly zero.

7 Open-Source Ecosystem and Resource Deliverables

To support reproducible research, Perron is released as an open-source software and evaluation ecosystem:

- `scores.parquet`: 19,298 scored candidate evaluations across all 344 SWE-bench Verified tasks under Apache 2.0 (https://github.com/yvliet/perron).
- **`perron-core` v0.3.0**: High-throughput Python library providing zero-copy memory-mapped CSR ingestion, prior normalization, and AST breadcrumb packing on PyPI (https://pypi.org/project/perron-core/).
- `perron-mcp`: JSON-RPC stdio server for desktop developer agent environments.
- `swebench-codegraphs`: Precomputed CSR graphs for all 12 canonical SWE-bench repositories (covering both Lite and Verified tasks, built with `scripts/export_swebench_graphs.py`).
- `swebench-strata-annotations`: 800 lexical accessibility annotations across SWE-bench repositories (344 Verified evaluation tasks, CC0-1.0).
- **Kaggle Demonstration Workstation**: Interactive reproduction notebook executing end-to-end on free Dual Tesla T4 GPUs in 259.8 seconds (4.3 minutes), validating the 344-task context packing and 131-task Stratum C positive control at https://www.kaggle.com/code/yvliet/perron-gemma-4-workstation.

8 Conclusion

On SWE-bench Verified, call-graph 1-hop expansion does not demonstrate a statistically detectable advantage over matched lexical pools under a frozen Gemma 4 scorer, nor does it expand candidate reachability beyond lexical depth. While graph expansion adds 7.3 pp of candidate pool recall over seeds alone, the small model converts only $\sim$ 1 pp of it into top-5 accuracy. Matched pool truncation curves show that expanding candidate pool breadth under pointwise scoring yields flat or declining localization, consistent with distractor accumulation; in un-named defect tasks, neither expansion beats seeds alone or unscored BM25. Upstream boundary loss (41.6 pp) exceeds downstream compaction loss (11.3 pp) by a factor of 3.68× at $k = 5$ (1.2× at $k = 1$, 6.8× at $k = 10$). While point estimates nominally favor graph expansion (+0.87 to +2.03 pp), gains are statistically undetectable in 344 tasks and do not justify a 2.45× compute increase ($4,988$ extra forward passes over seeds; 5× pool cap). For local developer agents under tight compute budgets, retrieval engineering is more safely directed toward high-precision first-stage lexical retrieval and compact AST breadcrumb packing; exploring whether listwise reranking architectures can overcome distractor accumulation remains a key direction for future research.

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A Linear Algebraic Formulations and Theoretical Proofs

A.1 Brauer Rank-1 Spectral Bound and Second Eigenvalue

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a directed multigraph and $T \in \mathbb{R}^{|\mathcal{V}|\times|\mathcal{V}|}$ be the row-substochastic transition matrix defined in Equation 2. Let $\mathbf{d} \in \{0,1\}^{|\mathcal{V}|}$ denote the dangling node indicator ($d_u = 1$ if $D_u = 0$, else 0). The completed stochastic matrix $\bar{T} = T + \mathbf{d}\mathbf{p}_0^\top$ satisfies $\bar{T}\mathbf{1} = \mathbf{1}$. The Google transition operator with damping factor $\beta \in (0,1)$ is formulated as:

$$M = \beta\bar{T} + (1-\beta)\mathbf{1}\mathbf{p}_0^\top. \tag{7}$$

By the Brauer rank-1 perturbation theorem applied to Markov transition matrices [6]: Let A be an $n \times n$ matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, and let $\mathbf{x}_1$ be the right eigenvector associated with λ_1. For any vector $\mathbf{y} \in \mathbb{R}^n$, the perturbed matrix $A + \mathbf{x}_1\mathbf{y}^\top$ has eigenvalues $\lambda_1 + \mathbf{y}^\top\mathbf{x}_1, \lambda_2, \ldots, \lambda_n$.

In our formulation, setting $A = \beta\bar{T}$ yields eigenvalues $\beta\lambda_1(\bar{T}), \beta\lambda_2(\bar{T}), \ldots, \beta\lambda_n(\bar{T})$. Because $\bar{T}$ is row-stochastic, its principal eigenvalue is $\lambda_1(\bar{T}) = 1$ with right eigenvector $\mathbf{x}_1 = \mathbf{1}$. Setting $\mathbf{y} = (1-\beta)\mathbf{p}_0$ gives:

$$M = \beta\bar{T} + \mathbf{1}((1-\beta)\mathbf{p}_0)^\top. \tag{8}$$

The principal eigenvalue of M becomes:

$$\lambda_1(M) = \beta\lambda_1(\bar{T}) + ((1-\beta)\mathbf{p}_0)^\top\mathbf{1} = \beta + (1-\beta) = 1. \tag{9}$$

All remaining eigenvalues satisfy:

$$\lambda_i(M) = \beta\lambda_i(\bar{T}), \quad \forall i \geq 2. \tag{10}$$

Because $\bar{T}$ is row-stochastic, the Gershgorin circle theorem guarantees $|\lambda_i(\bar{T})| \leq 1$ for all i. Therefore, the second-largest eigenvalue is strictly bounded:

$$|\lambda_2(M)| \leq \beta \cdot 1.0 = \beta. \tag{11}$$

For $\beta = 0.85$, the spectral gap is uniformly lower-bounded:

$$1 - |\lambda_2(M)| \geq 1 - \beta = 0.15, \tag{12}$$

guaranteeing geometric convergence $O(\beta^t)$ in L_1 total variation distance regardless of graph density or disconnected components.

A.2 Barycentric Linearity and Basis Warping Bounds

Following Jeh and Widom [7], let $\mathbf{p}_0 = \sum_{i=1}^K w_i\mathbf{e}_{s_i}$ be any convex combination of query seed nodes with $w_i \geq 0$ and $\sum w_i = 1.0$. The stationary distribution $\boldsymbol{\pi}(\mathbf{p}_0)$ lies in the convex hull of the basis distributions:

$$\boldsymbol{\pi}(\mathbf{p}_0) = \sum_{i=1}^K \tilde{w}_i\boldsymbol{\pi}(\mathbf{e}_{s_i}) \in \operatorname{conv}\left(\{\boldsymbol{\pi}(\mathbf{e}_{s_i})\}_{i=1}^K\right), \quad \tilde{w}_i = \frac{w_i m_i}{\mathbf{w}^\top\mathbf{m}}, \tag{13}$$

where basis mass satisfies $m_i = \|(1-\beta)\mathbf{e}_{s_i}(I-\beta T)^{-1}\|_1 \in [1-\beta, 1.0]$. The maximum coordinate warping ratio is bounded by:

$$\frac{\max_i m_i}{\min_j m_j} \leq \frac{1}{1-\beta} = \frac{1}{0.15} \approx 6.67. \tag{14}$$

A.3 Markov Renewal Equivalence to Normalized Leaky Resolvents

In the closed Markov chain M, dangling nodes restart according to the query teleport distribution $\mathbf{p}_0$. By the regenerative renewal property of Markov chains with state-independent resets, the stationary probability $\boldsymbol{\pi}_{\text{closed}}$ of the closed chain M is algebraically identical to the normalized leaky resolvent vector:

$$\boldsymbol{\pi}_{\text{leaky}} = \frac{(1-\beta)\mathbf{p}_0(I-\beta T)^{-1}}{\|(1-\beta)\mathbf{p}_0(I-\beta T)^{-1}\|_1}. \tag{15}$$

Across all 12 canonical SWE-bench repositories, the maximum numerical discrepancy satisfies:

$$\|\boldsymbol{\pi}_{\text{closed}} - \boldsymbol{\pi}_{\text{leaky}}\|_1 = 1.68 \times 10^{-8} < 2 \times 10^{-8}, \tag{16}$$

confirming exact algebraic equivalence.

A.4 Asymptotic Pointwise Mutual Information (PMI) Decomposition

For any code symbol v with positive query support, the log-transformed Specificity score satisfies:

$$\ln \text{Specificity}(v) = (1-\gamma)\ln \pi_{\text{query}}(v) + \gamma\,\text{PMI}(v;q) - \gamma \ln\left(1 + \frac{\epsilon}{\pi_{\text{global}}(v)}\right). \tag{17}$$

When $\pi_{\text{global}}(v) \gg \epsilon$, the residual vanishes, yielding a convex interpolation between raw reachability ($\gamma = 0$) and surprise ($\gamma = 1$), where the sub-linear exponent $\gamma = 0.70$ acts as a structural Inverse Document Frequency (IDF) discount.

A.5 Neumann Series Expansion and Sink Absorption

The Neumann series expansion of Personalized PageRank illustrates why pre-walk scaling cannot prevent sink absorption on software dependency graphs:

$$\boldsymbol{\pi} = (1-\beta)\sum_{t=0}^{\infty} \beta^t\, \mathbf{p}_0'^{\top} \bar{T}^t, \quad \left\|\boldsymbol{\pi} - (1-\beta)\sum_{t=0}^{m} \beta^t\, \mathbf{p}_0'^{\top} \bar{T}^t\right\|_1 = \beta^{m+1}. \tag{18}$$

Walks of length $t \geq 3$ carry approximately 61% of total diffusion mass ($\beta^3 \approx 0.614$ at $\beta = 0.85$). Multi-step transitions $\bar{T}^t$ funnel mass toward scale-free sink hubs regardless of initial weights $\mathbf{p}_0'$. Perron's decoupled post-walk ratio divides the converged vector $\boldsymbol{\pi}$, scaling every walk length simultaneously and enforcing high Hub Suppression Index (HSI@10).

B Perron CSR SpMM Engine Architecture and Systems Benchmarks

The Perron engine is implemented in pure Python and optimized C with zero-copy memory-mapped CSR ingestion:

- **Ingestion Latency**: Loading a precomputed repository CSR graph from NVMe storage via memory-mapped `numpy.load` takes < 1.2 ms across all repositories.
- **Memory Footprint**: Total resident set size (RSS) during SpMM diffusion is bounded below 144 MB RAM even on SymPy ($|\mathcal{V}| = 37{,}442$, $|\mathcal{E}| = 148{,}912$).
- **Throughput Comparison**: Compared to standard NetworkX PageRank (320 ms on Requests) and Scipy sparse iterative solvers (28 ms), Perron executes power iteration in 1.89 ms on Requests and 5.10 ms on SymPy, achieving a $> 50\times$ speedup suitable for interactive developer agent tool loops.

C Continuous Dose-Response Surface Across 186 Prior Configurations

To evaluate the entropy collapse hypothesis systematically, a continuous resolvent basis benchmark was evaluated across the 344 Verified evaluation tasks (63,984 SpMM solves) across four parametric families:

1. **Softmax Family**: $\mathbf{p}_0(i) \propto \exp(s_i/\tau)$ for top-K seeds ($\tau \in [0.05, 1.0]$).
2. **Power Family**: $\mathbf{p}_0(i) \propto s_i^p$ ($p \in [0.5, 4.0]$).
3. **Linear Truncated**: $\mathbf{p}_0(i) \propto s_i$ over top-K seeds ($K \in [5, 100]$).
4. **Rank-Zeta Family**: $\mathbf{p}_0(i) \propto i^{-\alpha}$ ($\alpha \in [0.2, 2.0]$).

Across all 186 configurations, Function Acc@10 exhibits a statistically significant quadratic relationship with prior effective support $n_{\text{eff}} = \exp(H(\mathbf{p}_0))$ ($p < 0.001$, binomial GLMM $R^2 = 0.197$), peaking near $n^*_{\text{eff}} \approx 8.8$ at 50.9%, which exactly ties Okapi BM25 (50.9%).

D Lexical Accessibility Annotation Protocol

Tasks are categorized into three mutually exclusive strata:

- **Stratum A (Named, $N = 208$, 60.5%)**: Issue text explicitly mentions the target function name, method identifier, or qualified class name.
- **Stratum B (Traceback, $N = 5$, 1.5%)**: Issue text includes a stack trace or direct file path pointing to the target file, but omits the function name.
- **Stratum C (Un-Named, $N = 131$, 38.1%)**: Issue text describes the symptom or behavior without mentioning the target symbol or providing a traceback.

Annotation was validated via an independent 50-task spot check with 100% inter-rater agreement and is released in `swebench-strata-annotations` (CC0-1.0).